

Machine Learning II
Group Assignment
Spotify Hit Song Prediction

Section 1 Group 5:

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Executive Summary

This report is designed to support the music and record label companies in identifying songs with the potential to become hit tracks, enabling more strategic allocation of resources and marketing budgets. The music industry operates on a hits-driven model, where a small number of songs generate a disproportionate share of revenue. This makes early and accurate identification of high-potential tracks critical to maximising commercial success. Wrongly identifying hit songs can be costly due to delayed or lost revenue – for example “*Truth Hurts*” by Lizzo and “*Die For You*” by The Weeknd only achieved mainstream success years after their initial release. These cases highlight the cost of missed opportunities, where songs with strong potential are not fully supported early on, resulting in delayed or lost revenue.

The analysis is based on Spotify data sourced from Kaggle, covering over 114,000 tracks and 21 descriptive features. Several machine learning classification models were evaluated, with the final model selected based on its ability to effectively identify high-potential songs and support business decision-making.

The findings indicate that **audio characteristics play a more significant role than genre** in determining a song’s success. In particular, features such as **loudness, danceability, and vocal presence (low instrumental content)** are strong indicators of hit potential.

Based on these insights, three key recommendations are proposed:

1. **Prioritise early investment in high-potential tracks** to avoid missing breakout opportunities
2. **Adopt a tiered marketing strategy** to balance opportunity capture with cost control
3. **Leverage audio insights in production and marketing decisions** to align creative output with commercial success factors

Overall, this report provides a data-driven framework to support more informed investment decisions, helping record labels maximise returns while reducing the risk of overlooking high-value opportunities.

1. Data Audit and Exploratory Data Analysis

The dataset comprised of 114,000 Spotify tracks with a total of 21 descriptive variables – combining both genres and audio attributes such as danceability, energy and loudness (Refer to Annex A for full description of variables). The data also consists of a popularity score (between 0 and 100) which will reflect a higher score if the song is played more times on Spotify.

Before performing any modelling, it is essential that the data is audited to ensure business reliability. Checks on full duplicate rows were first performed followed by tracks and artist combination, which resulted in a removal of 450 and 32,000 rows respectively. A check on the missing values was also performed which resulted in the removal of 1 row. Finally, the datatype of variables was fixed to their appropriate Boolean and Categorical datatypes, resulting in a final clean dataset of 81,343 rows. Feature engineering was then performed to create more insightful analysis such as mood_intensity which was the combination of valence and energy (Refer to Annex A for detailed explanation of variables created).

Exploratory Data Analysis (EDA) was conducted next to understand the distribution of the cleaned data. An analysis of the categorical variables (track_genre) revealed a total of 114 unique values which will not be feasible for modelling purposes. As such, a genre_group was

created to group similar genre values together such as 'metal' and 'heavy-metal' are both grouped under the 'metal' genre group. The resulting genre_group consists of a total of 12 categories, to enable better analysis of features.

2. Classification Models

2.1 Technical Preparation Before Modelling

Before any model was trained, several steps were made to prepare the data appropriately. A song was defined as a hit if its popularity score was above the 75th percentile of the distribution, capturing the most successful tracks in the dataset. The popularity feature was immediately dropped to prevent data leakage.

Nine final characteristics were selected as predictors, eight of which were audio characteristics (loudness, acousticness, instrumentalness, liveness, danceability, energy, valence and the engineered mood intensity) and variable genre group (consolidation of 114 genres into 15 meaningful ones). Features with no meaningful relationship to success were discarded. The data was divided into a training set (70%) to build models, and a test set (30%) to evaluate the performance of the model. Finally, three models were developed and compared.

2.2 Exploring the models

A Decision Tree was used as a baseline due to its interpretability, but it showed clear signs of overfitting and weak generalisation performance. A Random Forest model improved results by reducing variance through bootstrapping and random feature selection, leading to more stable predictions. However, its performance remained limited in capturing more complex relationships in the data.

Finally, a Boosting model (XGBoost) was applied. Unlike Random Forest, Boosting trains trees sequentially, focusing on correcting previous errors. This allows it to better capture complex patterns and interactions. The results show that Boosting outperforms both Decision Tree and Random Forest across all key metrics, particularly in F1-score and ROC-AUC, making it the most effective model for this problem.

2.3 Final Classification Model

Boosting (XGBoost) was selected as the final model as it outperformed Decision Tree and Random Forest across all key metrics, particularly F1-score and ROC-AUC. Its sequential learning approach allows it to better capture complex patterns by correcting previous errors. With proper tuning, it provides the best balance between performance and generalisation.

3. Business Insights

Following the selection of XGBoost as the final model, a deeper analysis was conducted to identify the key drivers of song success and translate these into actionable business insights. 3 key insights were identified and can be summarised as such:

1. Audio characteristics > genre
2. Missing potential hits has the greatest financial impact
3. Capturing more potential hits increases overall returns, albeit earlier inefficiency

3.1 Understanding the key drivers of a hit song

To further understand the key drivers of hit songs, both permutation importance and SHAP analysis were conducted. Results from both methods consistently display audio features such as loudness, acousticness and energy as the most influential predictors.

From the SHAP beeswarm plot, the direction of impact can also be observed

- Higher loudness significantly increase the probability of a song becoming a hit
- Lower instrumentalness (i.e. more vocal-driven tracks) is associated with higher success

- Greater danceability positively contributes to hit potential
- Higher acousticness reduces the likelihood of a song being a hit

These findings indicate that **audio characteristics are the primary drivers of success**, while genre plays a secondary role.

3.2 Translating to economic insights

To illustrate the real business impact of our analysis, a profit-based framework was applied using simulated monetary values.

In the music industry, a small number of hit songs generate a disproportionate share of revenue.

- **False Negatives (missing a hit)** → high opportunity cost
- **False Positives (promoting a non-hit)** → lower financial cost

Missing a hit therefore represents a significant lost opportunity—not only in terms of streams, but also branding, audience growth, and long-term artist value. In contrast, promoting a song that does not become a hit incurs a financial cost, but with relatively lower long-term impact. As such, the key priority is to **maximise recall**, ensuring that potential hit songs are not overlooked.

To reflect this, a profit-based threshold optimisation was conducted using the following assumptions:

- €10,000 gain from correctly identifying a hit
- €5,000 loss from missing a hit
- €1,000 cost of promoting a non-hit.

The analysis identifies an optimal decision threshold of **0.23**, which is significantly lower than the conventional 0.5. This indicates that a more aggressive strategy in identifying potential hits leads to higher overall financial returns. At this threshold, the model delivers an expected profit of **€1,821 per case** on the test set, while achieving a **recall of 0.97**, meaning that 97% of actual hit songs are successfully identified. The confusion matrix further shows that **missed hits are minimal (0.67%)**, significantly reducing the risk of lost high-value opportunities.

This performance involves a trade-off in precision (0.29), meaning that some songs predicted as hits do not ultimately succeed, resulting in a higher proportion of false positives. As shown in the confusion matrix (Figure B16), this reflects a larger number of non-hit songs being incorrectly promoted. However, this is economically justified, as the cost of missing a hit significantly outweighs the cost of promoting a non-hit.

The results support a high-recall, opportunity-focused strategy, where maximising the detection of potential hits leads to higher overall financial returns, despite some inefficiency in resource allocation. The profit-based threshold optimisation therefore provides a practical decision tool, enabling record labels to prioritise high-potential songs while minimising the risk of missing valuable opportunities.

4. Recommendations, Conclusions, Limitations

Based on the business insights gathered in Part 3, three key recommendations are proposed to support record labels in making more effective and data-driven investment decisions.

4.1 Prioritise Early Investment in High-Potential Tracks

The model enables record labels to identify songs with a higher likelihood of becoming hits based on their audio characteristics and genre. As such, it should be used as an **early**

decision-support tool to guide how marketing budgets are allocated across upcoming releases.

The key objective is to **avoid missing potential breakout songs**, as these generate significant revenue, brand exposure, and long-term artist value. By casting a wider net and identifying more potential hits early, labels can ensure that high-value opportunities are not overlooked. While this approach may result in promoting some songs that do not ultimately succeed, the cost of doing so is relatively small compared to the impact of missing a true hit.

4.2 Implement a Tiered Marketing Strategy to Manage Risk

Given that not all selected songs will succeed, record labels should adopt a **tiered and adaptive marketing strategy**:

- Allocate initial budgets across a broader pool of promising tracks
- Gradually increase investment in songs that demonstrate early traction (e.g., strong streaming growth, social engagement)

This approach allows labels to remain agile and responsive to market signals, while balancing **opportunity capture with cost control**. It ensures that resources are concentrated on songs that show real momentum, without risking missed breakout successes at the early stage.

4.3 Leverage Audio Features in Production and Marketing Decisions

The analysis shows that **audio features are more important than its genre alone** in determining commercial success. In particular, characteristics such as loudness, danceability, and strong vocal presence are consistently associated with higher-performing tracks.

Record labels and producers should incorporate these insights into the production and A&R process by:

- Prioritising these characteristics during song development and mixing
- Using them as early signals when selecting songs for release
- Aligning promotional strategies with tracks that exhibit these attributes

This ensures that creative decisions are better aligned with what drives audience engagement and commercial performance.

4.4 Limitations

Our analysis focuses primarily on audio features and genre to support early identification of potential hits and hence early-stage investment decisions. However, there exist several limitations such as:

- Influence of Artist is not factored in the analysis: Fan base and strength of personal branding are relevant factors that affect the song's performance
- Changing market trends are not captured: Musical preferences and styles evolve over time and changes between decades
- Impact of social media and virality are not considered: These are recent factors that play an increasingly important role in driving a song's success

Ultimately, this analysis is based on Spotify data only and may not capture the full performance of a song across all streaming platforms or channels.

4.5 Overall Conclusion

Overall, a data-driven and opportunity-focused approach enables record labels to maximise returns by identifying high-potential songs early and allocating investment efficiently, minimising missed opportunities.

While marketing investment influences a song's success, labels must first decide which songs to support. The model addresses this initial selection problem by identifying high-potential tracks prior to resource allocation.

TECHNICAL ANNEX A – DATA MANIPULATION AND EXPLORATION

This annex supports Section 1 of the main report and aligns with the technical Python notebook. It provides further detail on the data audit, data cleaning, feature engineering, and exploratory analysis conducted before classification modelling

A1.1 Data Exploration

Initial exploratory analysis of the data revealed several data quality issues.

	Unnamed: 0	popularity	duration_ms	danceability	energy	key	loudness	
count	114000.000	114000.000	114000.000	114000.000	114000.000	114000.000	114000.000	
mean	56999.500	33.239	228029.153	0.567	0.641	5.309	-8.259	
std	32909.110	22.305	107297.713	0.174	0.252	3.560	5.029	
min	0.000	0.000	0.000	0.000	0.000	0.000	-49.531	
25%	28499.750	17.000	174066.000	0.456	0.472	2.000	-10.013	
50%	56999.500	35.000	212906.000	0.580	0.685	5.000	-7.004	
75%	85499.250	50.000	261506.000	0.695	0.854	8.000	-5.003	
max	113999.000	100.000	5237295.000	0.985	1.000	11.000	4.532	
	mode	speechiness	acousticness	instrumentalness	liveness	valence	tempo	time_signature
114000.000	114000.000	114000.000	114000.000	114000.000	114000.000	114000.000	114000.000	114000.000
	0.638	0.085	0.315	0.156	0.214	0.474	122.148	3.904
	0.481	0.106	0.333	0.310	0.190	0.259	29.978	0.433
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.036	0.017	0.000	0.098	0.260	99.219	4.000
	1.000	0.049	0.169	0.000	0.132	0.464	122.017	4.000
	1.000	0.085	0.598	0.049	0.273	0.683	140.071	4.000
	1.000	0.965	0.996	1.000	1.000	0.995	243.372	5.000

Figure A1: Summary of Descriptive Statistics of Original Data

Figure A1 presents the descriptive statistics of the numerical variables in the original dataset, confirming that the database contains 114,000 observations. The summary provides an initial overview of the central tendency, dispersion, and range of the main audio features and popularity metric. In particular, the average song popularity is 33.24, the mean duration is approximately 228,029 milliseconds (around 3.8 minutes), and the average tempo is 122.15 BPM. The table also highlights the existence of extreme values in variables such as duration, loudness, and tempo, which justified a more detailed exploration and later outlier treatment during the data cleaning stage.

To fully understand the data at hand, it is important to establish definitions of the variables present in the dataset. These are the definitions that will be used through the project:

Column	Explanation
track_id	Unique Spotify ID assigned to each track.
artists	Name of the artist or artists who performed the song.
album_name	Name of the album in which the track appears.
track_name	Title of the song.
popularity	Spotify popularity score from 0 to 100, indicating how popular a track is on the platform.
duration_ms	Length of the track measured in milliseconds.

explicit	Indicates whether the song contains explicit lyrics.
danceability	Measure of how suitable a track is for dancing based on rhythm, tempo, and beat stability.
energy	Measure of intensity and activity; high values represent songs that feel fast, loud, and energetic.
key	Musical key in which the track is composed.
loudness	Overall loudness of the track, measured in decibels (dB).
mode	Indicates whether the song is in major mode or minor mode.
speechiness	Measure of the presence of spoken words in the track.
acousticness	Confidence measure of whether the track is acoustic.
instrumentalness	Measure of the likelihood that the track contains no vocals.
liveness	Measure of the probability that the track was performed live.
valence	Measure of the musical positivity of the track; high values sound happier or more cheerful.
tempo	Estimated speed of the track in beats per minute (BPM).
time_signature	Number of beats in each musical bar of the track.
track_genre	Genre label assigned to the track.

Figure A2: Summary of Descriptive Statistics of Original Data

Figure A2 presents the variables included in the original Spotify dataset together with a short description of each one. The dataset combines identification fields, song and album information, genre classification, and several numerical audio attributes that describe musical characteristics such as danceability, energy, loudness, tempo, and valence. This overview is useful for understanding the role of each variable before proceeding with the data cleaning, feature engineering, and modelling stages.

A1.2 Data Cleaning

The first cleaning step was dropping the unnamed column since it was a redundant indexer. The dataset was then checked for duplicate observations, revealing 450 fully duplicated rows out of 114,000 total records, which were removed to reduce the dataset to 113,550 rows. Next, repeated track name and artist combinations were examined, as the same song could appear multiple times under different genre labels.

```

Rows before deduplication: 113,550
Rows removed:              32,206
Rows after deduplication:  81,344

```

Figure A3: Duplicate Song Reduction

To address this, the data was sorted by popularity in descending order and duplicates were removed while keeping only the most popular entry for each track-artist pair. This second deduplication step shown on Figure A3 reduced the dataset from 113,550 to 81,344 rows, removing 32,206 observations and ensuring that each song was represented only once in the final cleaned dataset. Also, there was only one row with missing values, since removing this instance will not cause significant data loss, it was removed from the dataset.

```
<class 'pandas.core.frame.DataFrame'>
Index: 81343 entries, 0 to 81343
Data columns (total 20 columns):
#   Column              Non-Null Count  Dtype
---  -
0   track_id            81343 non-null  object
1   artists             81343 non-null  object
2   album_name          81343 non-null  object
3   track_name          81343 non-null  object
4   popularity           81343 non-null  int64
5   duration_ms         81343 non-null  int64
6   explicit            81343 non-null  bool
7   danceability         81343 non-null  float64
8   energy              81343 non-null  float64
9   key                 81343 non-null  int64
10  loudness            81343 non-null  float64
11  mode                81343 non-null  bool
12  speechiness         81343 non-null  float64
13  acousticness        81343 non-null  float64
14  instrumentalness     81343 non-null  float64
15  liveness            81343 non-null  float64
16  valence              81343 non-null  float64
17  tempo               81343 non-null  float64
18  time_signature       81343 non-null  int64
19  track_genre         81343 non-null  category
dtypes: bool(2), category(1), float64(9), int64(4), object(4)
memory usage: 11.4+ MB
```

Figure A4: Final Data Types

After removing duplicates, the data types of selected variables were adjusted to better reflect their meaning and improve later analysis. The variable mode was converted to a Boolean format, since it only indicates whether a track is in major or minor mode, while track_genre was converted to a categorical variable to better represent genre labels and reduce memory usage. Figure A4 presents the final structure of the cleaned dataset, including all variables and their corresponding data types after these adjustments.

	popularity	duration_ms	danceability	energy	key	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo	time_signature
count	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000	81343.000
mean	35.249	231400.358	0.559	0.635	5.285	-8.599	0.089	0.330	0.185	0.220	0.463	122.131	3.897
std	19.410	116484.219	0.178	0.259	3.557	5.306	0.117	0.340	0.332	0.198	0.263	30.123	0.457
min	0.000	8586.000	0.000	0.000	0.000	-49.531	0.000	0.000	0.000	0.000	0.000	0.000	0.000
25%	21.000	173862.000	0.446	0.455	2.000	-10.459	0.036	0.016	0.000	0.099	0.241	99.368	4.000
50%	35.000	215200.000	0.573	0.678	5.000	-7.267	0.049	0.190	0.000	0.133	0.449	122.028	4.000
75%	49.000	267333.000	0.690	0.857	8.000	-5.143	0.087	0.629	0.153	0.283	0.676	140.123	4.000
max	100.000	5237295.000	0.985	1.000	11.000	4.532	0.965	0.996	1.000	1.000	0.995	243.372	5.000

Figure A5: Data Set After Cleaning

Figure A5 presents the summary statistics of the dataset after the cleaning process. Following the removal of duplicate observations and repeated track-artist entries, the final cleaned dataset contained 81,343 rows. The table shows the distribution of the main numerical variables and confirms that the cleaning process preserved the overall structure of the data while reducing redundancy. It also provides an updated view of central tendency, dispersion, and extreme values, which served as the basis for the subsequent exploratory analysis.

A1.3 Features Engineering

To support a more meaningful analysis, several new variables were created from the cleaned dataset. These new variables are shown on Figure A6:

	duration_min	num_artists	is_collab	energy_danceability	mood_intensity
0	2.616	2	True	0.337	0.112
1	3.316	2	True	0.486	0.430
2	2.921	2	True	0.541	0.293
3	2.711	1	False	0.567	0.577
4	2.976	2	True	0.649	0.303

Figure A6: New Calculated Features

Track duration was converted from milliseconds into minutes to make interpretation easier, while the number of artists involved in each song was calculated to capture collaboration size. A separate Boolean variable was also created to indicate whether a track is a collaboration or a solo performance. In addition, two interaction variables were introduced: `energy_danceability`, which combines how energetic and danceable a song is, and `mood_intensity`, which combines valence and energy to reflect the overall emotional intensity of a track. These engineered features helped enrich the dataset with more interpretable and potentially predictive information for the later stages of the analysis.

To prepare the dataset for the next stages of the analysis, the variables were grouped according to their data type. This step was useful to distinguish numerical, Boolean, categorical, and text variables so that each group could later receive the appropriate exploratory analysis and preprocessing treatment.

```

Numerical:
Index(['popularity', 'duration_ms', 'danceability', 'energy', 'key',
      'loudness', 'mode', 'speechiness', 'acousticness', 'instrumentalness',
      'liveness', 'valence', 'tempo', 'time_signature', 'duration_min',
      'num_artists', 'energy_danceability', 'mood_intensity'],
      dtype='object')

Categorical:
Index([], dtype='object')

Boolean:
Index(['explicit', 'is_collab'], dtype='object')

Strings:
Index(['track_id', 'artists', 'album_name', 'track_name', 'track_genre'], dtype='object')

```

Figure A7: Grouping Variables by Data Type

As shown in the output, most variables were classified as numerical, including both the original Spotify audio features and the newly engineered variables. The Boolean variables were `explicit` and `is_collab`, while the text variables included identifiers and descriptive fields such as `track_id`, `artists`, `album_name`, `track_name`, and `track_genre`. This organization provided a clearer structure for the dataset and supported the following exploration steps.

A1.4 Exploratory Data Analysis

As part of the exploratory data analysis, the Boolean variables were examined to better understand their distribution in the cleaned dataset. This step helps identify whether these

variables are balanced or dominated by one category, which is useful for interpreting the data and anticipating their potential contribution to later modelling.

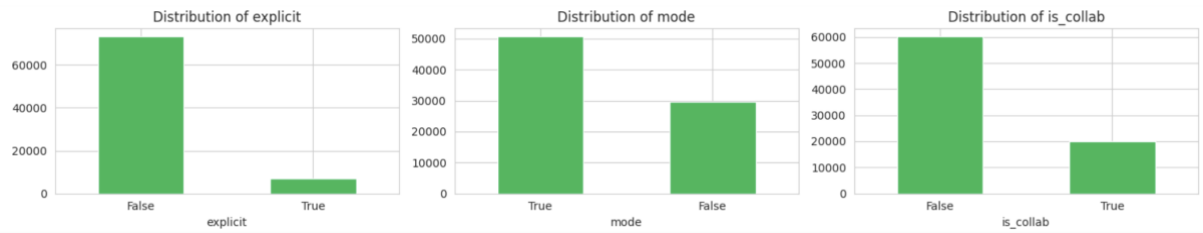


Figure A7: Grouping Variables by Data Type

Figure A7 shows that most songs in the dataset are not explicit, indicating that explicit content represents a smaller share of the observations. For mode, tracks in major mode appear more frequently than those in minor mode. Similarly, the is_collab variable shows that most songs are not collaborations, while collaborative tracks make up a smaller portion of the dataset. Overall, these distributions suggest that the Boolean variables are not evenly balanced and may capture distinct but less frequent song characteristics.

To better understand the composition of the dataset, the distribution of the most frequent music genres was explored. Since the dataset contains a large number of genre labels, this visualization helps identify which genres appear most often and whether the data is concentrated in only a few categories.

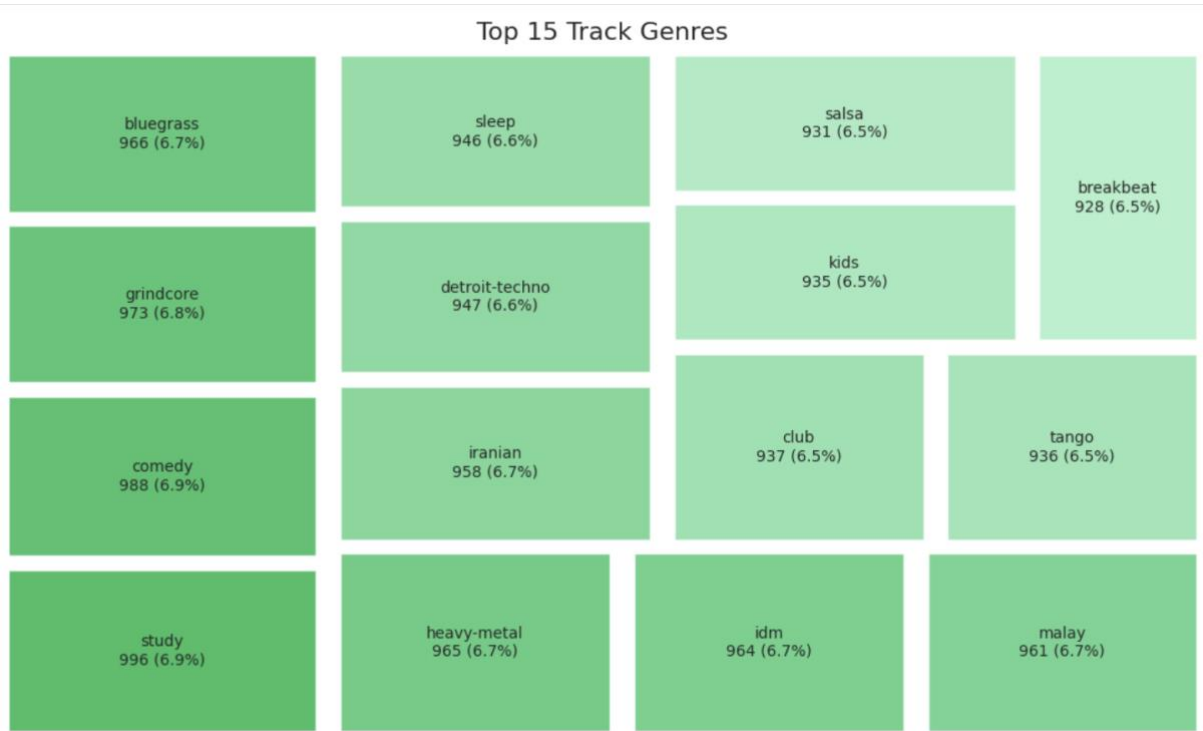


Figure A8: Treemap of the Top 15 Track Genres

This Figure A8 treemap shows that the top 15 genres have fairly similar frequencies, with no single genre dominating the dataset. Genres such as study, comedy, grindcore, bluegrass, and heavy-metal appear among the most frequent, but their shares remain close to one another, each representing roughly between 6.5% and 6.9% of the top-15 total. This suggests that the genre distribution is relatively dispersed, which supports the need for later grouping or consolidation of genre categories to simplify the analysis.

Because the dataset contains many detailed genre labels, these were grouped into broader categories to simplify the analysis and make the results easier to interpret.

CLASSICAL: classical, opera, piano	HIPHOP_RNB: gospel, hip-hop, r-n-b, soul	REGGAE_AFRO: afrobeat, dub, reggae, ska
DANCE: club, dance, dancehall, disco, j-dance, party	INDIE_ALT: alternative, indie	RELAX: ambient, chill, new-age, sleep, study

Figure A9: Examples of Grouping of Track Genres into Broader Categories

The output from Figure A9 shows that the original genre labels were consolidated into broader groups such as classical, dance, electronic, folk/country, hiphop/RnB, latin, metal, and other. This reduced the complexity of the track_genre variable and created a more practical structure for the following exploratory analysis and modelling stages.

The numerical variables were explored to better understand their distribution, central tendency, and potential irregularities in the cleaned dataset. This step is useful for identifying skewness, concentration patterns, and possible extreme values before moving into deeper analysis.

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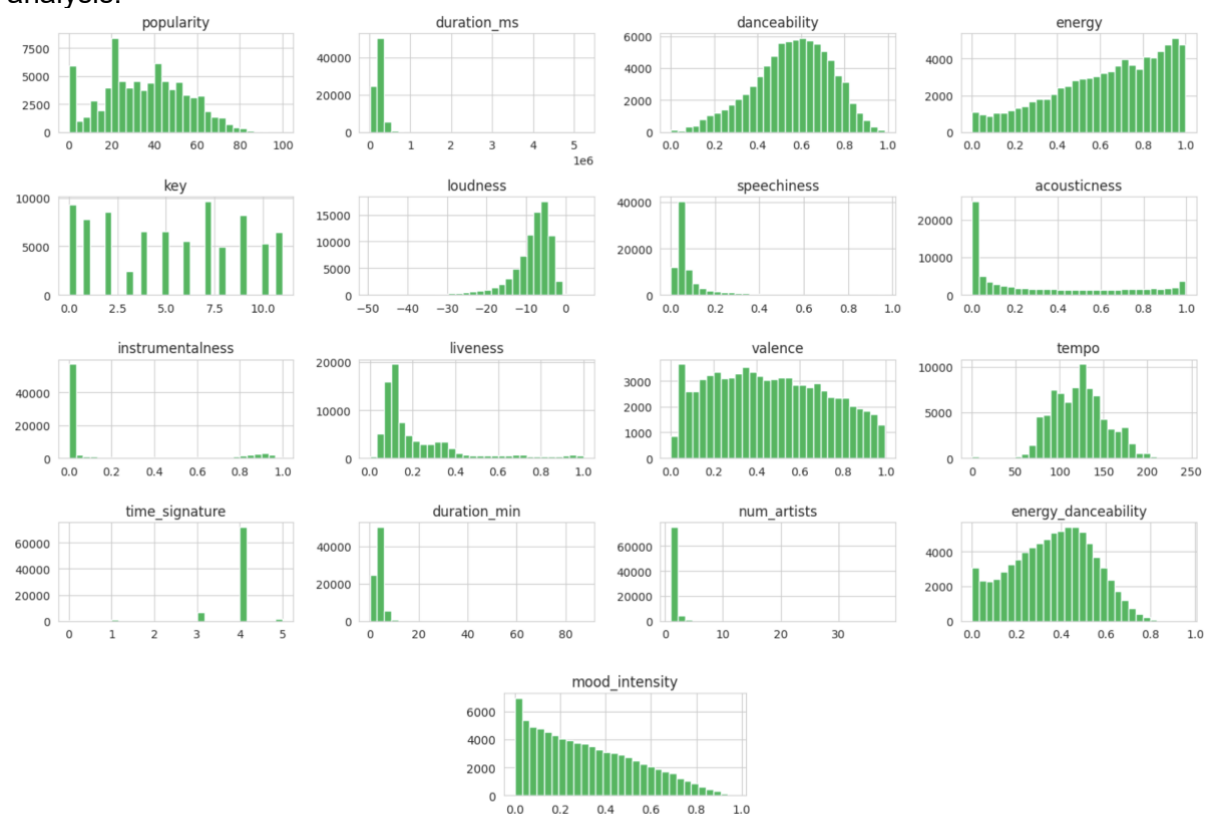


Figure A10: Histograms of Numerical Variables

The histograms in Figure A10 show that the numerical variables follow different distribution patterns across the dataset. Some variables, such as danceability, tempo, and energy_danceability, appear more concentrated around central ranges, while others such as duration_ms, duration_min, num_artists, speechiness, and instrumentalness are strongly right-skewed. Variables like time_signature and key are clearly discrete, while mood_intensity

and valence show broader spreads. Overall, the output confirms that the dataset contains a mix of symmetric, skewed, and bounded numerical features, which is important for interpreting later modelling results.

To complement the distribution analysis, boxplots were used to inspect the numerical variables for potential outliers and unusual observations. This step helps identify variables with extreme values that may influence the analysis and later modelling results.

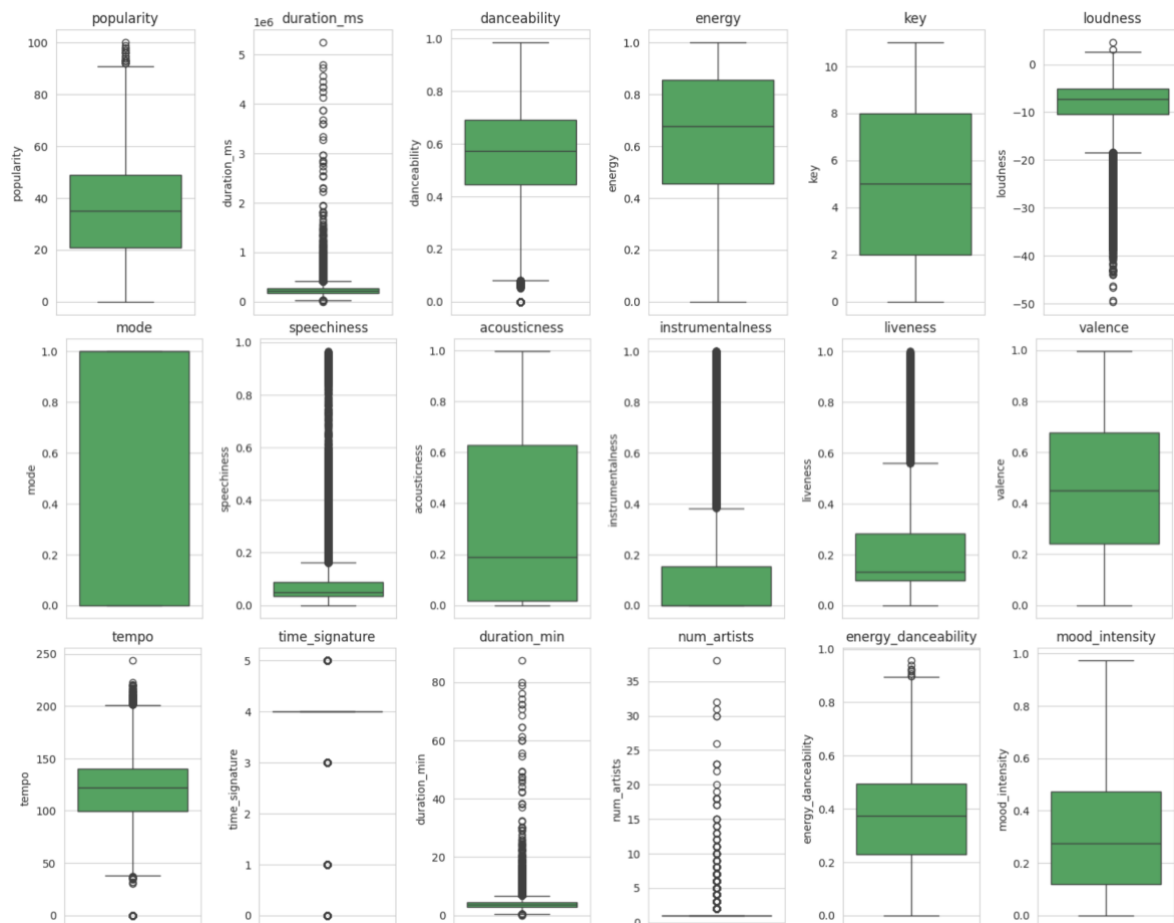


Figure A11: Boxplots of Numerical Variables

Figure A11 shows that several numerical variables contain noticeable outliers. In particular, duration_ms, duration_min, num_artists, speechiness, instrumentalness, liveness, and tempo display observations far from their central ranges. Variables such as danceability, energy, valence, energy_danceability, and mood_intensity appear more concentrated, while time_signature and key remain discrete by nature. Overall, the output confirms the presence of extreme values in multiple features, supporting the need to review these variables carefully in the data preparation process. Given the observations, extreme outliers were removed from three variables using simple business-based thresholds. Songs with a duration of 10 minutes or more were excluded because they are not typical of mainstream tracks. For tempo, only values between 40 and 200 BPM were kept, since values outside this range are likely to be unrealistic or caused by detection errors. Finally, tracks with loudness below -40 dB were removed because such values may indicate empty or near-silent audio. This cleaning step helped make the dataset more consistent and representative for the subsequent analysis.

A1.5 Target analysis

Before analysing the target distribution, a binary target variable called hit was created from the original popularity score. Since popularity is measured on a scale from 0 to 100, the 75th percentile was used as the threshold to distinguish commercially successful songs from the rest of the dataset. Songs with popularity values equal to or above this threshold were classified as hits, while the remaining songs were classified as non-hits. This approach provides a relative and data-driven definition of success that is suitable for classification modelling.

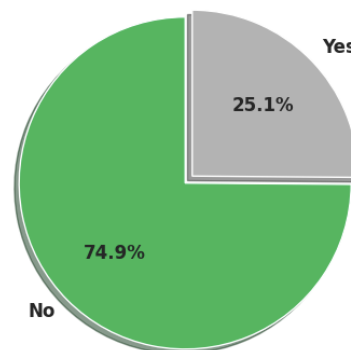
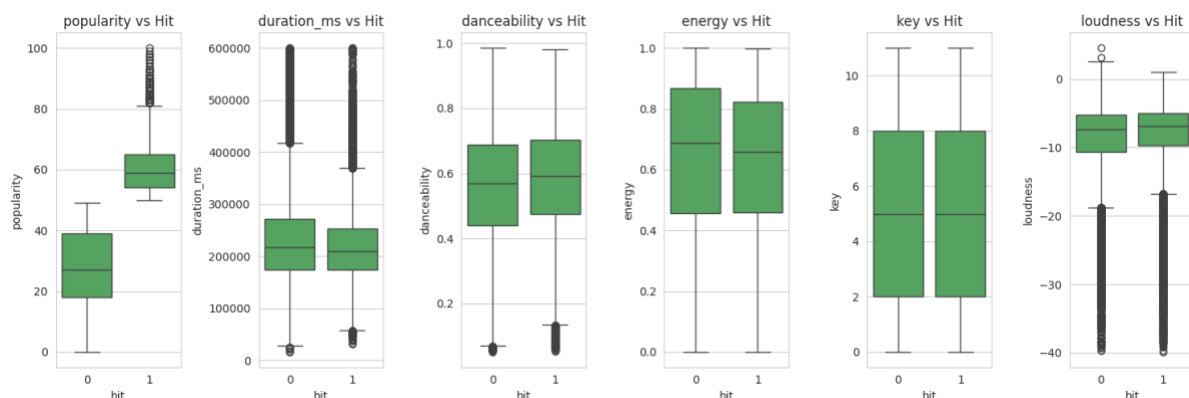


Figure A12: Distribution of the Target Variable (Hit vs Non-Hit)

As shown in Figure A13, the target variable is moderately imbalanced. Approximately 25.1% of the songs are classified as hits, while 74.9% are classified as non-hits. Although the classes are not perfectly balanced, the imbalance is not extreme, meaning the dataset remains suitable for standard classification modelling while still requiring attention during model evaluation.

To better understand how the target variable relates to the numerical features, boxplots were used to compare the distribution of each numerical variable between **hit** and **non-hit** songs. This helps identify which variables show noticeable differences across the two classes.



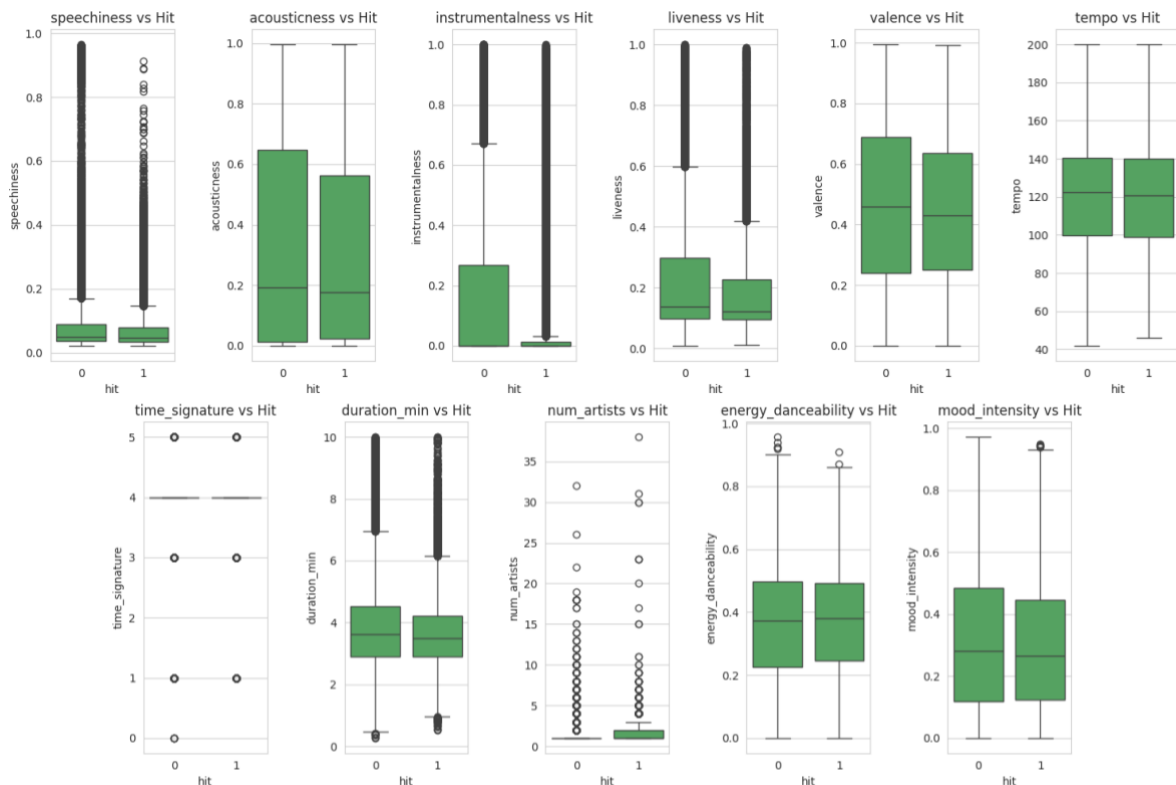


Figure A13: Boxplots of Numerical Variables by Target Class

The boxplots in Figure A13 suggest that some numerical variables show clearer differences between hit and non-hit songs than others. In particular, popularity, danceability, energy, and loudness appear to vary more visibly across the two groups, while variables such as key, tempo, duration, and time_signature show more similar distributions. Overall, the output indicates that certain audio features may be more useful than others for distinguishing hit songs from non-hit songs.

To examine the relationship between genre and song success, the hit rate was calculated for each grouped genre category. This allows a clearer comparison of which genre groups contain a higher proportion of hit songs in the dataset.

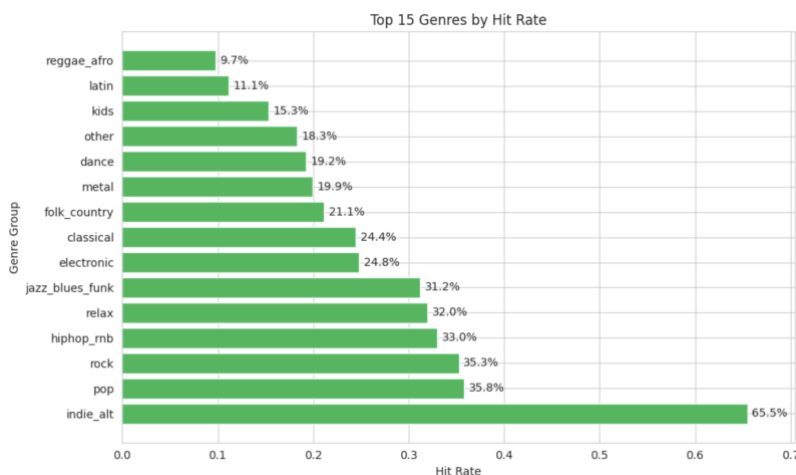


Figure A14: Genre Groups Ranked by Hit Rate

According to Figure A14, there are noticeable differences in hit rates across genre groups. Indie/alternative stands out with the highest hit rate at 65.5%, followed by pop and rock, which also show relatively high proportions of hit songs. In contrast, groups such as reggae/afro, latin, and kids display much lower hit rates. Overall, the output suggests that genre grouping may provide useful information for distinguishing between hit and non-hit songs.

TECHNICAL ANNEX B – CLASSIFICATION MODELS

This annex supports Section 2 of the main report and aligns with the technical Python notebook. It provides additional detail on the preprocessing, classification modelling, and model comparison steps carried out in the analysis.

Before presenting the individual models, this annex first outlines the data preparation decisions that made the classification stage possible. It explains how the target and predictor variables were defined, how the data was validated for modelling, and how the different classification approaches were structured and assessed before selecting the most appropriate solution.

B2.1 Target and Feature Selection

As it is necessary in for this classification model, the target variable and predictor set were defined. This step was necessary to ensure that the models were built using only the selected explanatory variables considered most relevant for predicting whether a song would be classified as a hit.

```
Columns kept: ['danceability', 'energy', 'loudness', 'acousticness', 'instrumentalness', 'liveness', 'valence', 'mood_intensity', 'genre_group', 'hit']
```

Figure B1: Columns Kept for Modeling

As shown in Figure B1, the target variable for the classification task was defined as hit, while the selected model features included a combination of key audio characteristics and engineered variables: danceability, energy, loudness, acousticness, instrumentalness, liveness, valence, mood_intensity, and genre_group. The output also confirms that only these selected predictors, together with the target variable, were retained in the modelling dataset.

Before fitting the first classification model, the dataset was validated to ensure that the modelling stage was based on a reliable train-test structure. This step was used to confirm that the target variable and selected predictors were ready for analysis and that the data split would allow model performance to be evaluated on unseen observations.

After validating the modelling dataset, a preprocessing pipeline was created to prepare the selected features for classification. Numerical variables were handled separately from categorical variables, allowing the appropriate transformations to be applied before model estimation. This step ensured that the training and test data were transformed consistently and provided a suitable input structure for the classification models.

B2.2 Classification Model – Option: 1 Classification Tree

As the first modelling approach, a Classification Tree was fitted to predict whether a song would be classified as a hit. The initial model was intentionally grown without depth restrictions in order to capture all possible splits in the training data, while using entropy as the splitting criterion and balanced class weights to account for the moderate class imbalance identified earlier.



DecisionTreeClassifier	
Parameters	
criterion	'entropy'
splitter	'best'
max_depth	None
min_samples_split	2
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	None
random_state	42
max_leaf_nodes	None
min_impurity_decrease	0.0
class_weight	'balanced'
ccp_alpha	0.0
monotonic_cst	None

Figure B2: Initial Configuration of the Classification Tree Model

As shown in Figure B2, the initial Classification Tree was built using entropy as the splitting criterion, a random state of 42 for reproducibility, and class_weight = "balanced" to reduce the effect of class imbalance during training. The model was left unconstrained at this stage, meaning that no maximum depth or pruning parameter was imposed initially. This provided a starting point for evaluating the full tree before later refinement and pruning.

```
recall of training set = 99.99%
recall of testing set = 38.00%
Number of nodes: 21609
Number of leaves: 10805
Depth: 58
```

Figure B3: Initial Performance of the Classification Tree

As shown in Figure B3, the initial Classification Tree achieved an almost perfect training recall of 99.99%, but only 38.00% recall on the testing set. The model also produced a very large structure, with 21,609 nodes, 10,805 leaves, and a depth of 58. These results strongly suggest that the tree was overfitting the training data, capturing too much detail from the sample and generalizing poorly to unseen observations. This confirmed the need for a pruning step to simplify the model and improve its out-of-sample performance.

To reduce the overfitting observed in the initial Classification Tree, a grid search procedure was applied to identify a more appropriate combination of tree depth and minimum leaf size. This tuning process aimed to simplify the model while preserving enough predictive power to classify hit songs effectively.

```
Fitting 5 folds for each of 12 candidates, totalling 60 fits
Best parameters found:
{'max_depth': 14, 'min_samples_leaf': 20}
Best F1 on training CV: 0.483
Number of leaves: 633
Depth: 14
Unique prob values: 335
```

Figure B4: Grid Search for the Optimal Tree Pruning Parameters

As shown in Figure B4, the Classification Tree was tuned using 5-fold cross-validation over different values of max_depth and min_samples_leaf, with the F1-score used as the selection criterion. The best parameter combination was max_depth = 14 and min_samples_leaf = 20. Compared with the initial unconstrained tree, this produced a much simpler model with 633

leaves and depth 14, indicating a substantial reduction in complexity and a more controlled structure for the final pruned tree.

To make the tuning results easier to interpret, the cross-validation scores were also visualised across the different combinations of `max_depth` and `min_samples_leaf`. This makes it possible to compare how model performance changes as the tree becomes more or less complex.

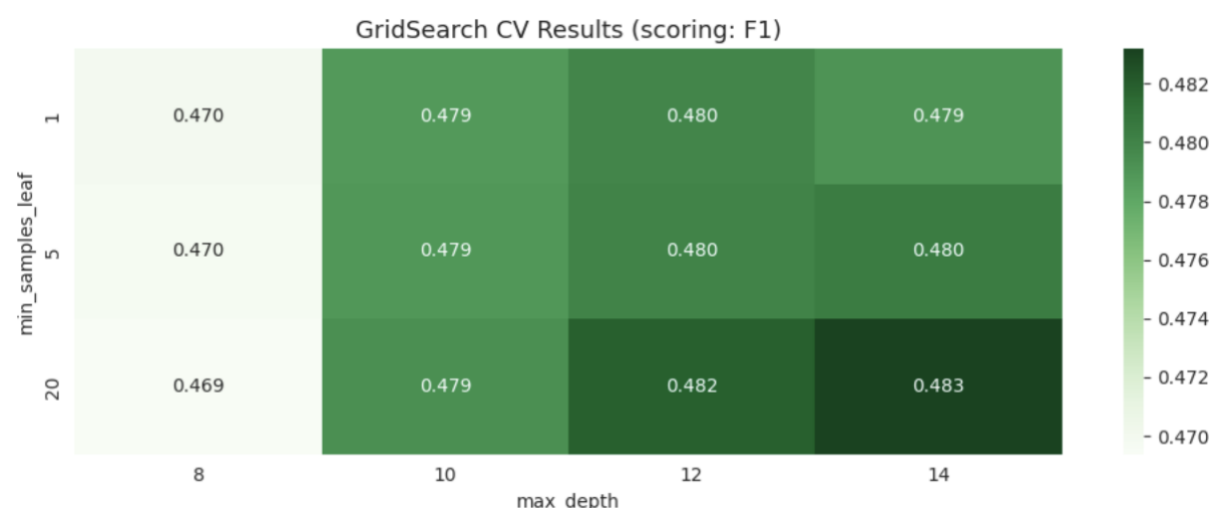


Figure B5: Heatmap of Grid Search CV Results for the Classification Tree

As shown in Figure B5, the best F1-score was achieved when the tree used a maximum depth of 14 and a minimum samples per leaf of 20. The heatmap also shows that performance generally improved as the tree depth increased from 8 to 14, while larger leaf sizes helped reduce overfitting and produced stronger validation results. Overall, Figure B5 confirms that the selected parameter combination offered the best balance between predictive performance and model complexity.

After tuning the tree structure, the next step was to determine the probability cutoff used to classify a song as a hit. Rather than relying automatically on the default threshold, the cutoff was selected by comparing precision and recall across different decision thresholds and choosing the value that maximized the F1-score.

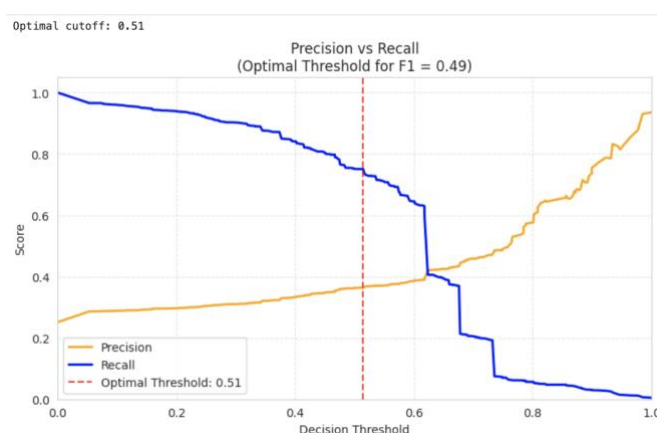


Figure B6: Precision-Recall Trade-off and Optimal Classification Threshold

Figure B6 plots the optimal decision threshold, which was identified at approximately 0.51, where the best balance between precision and recall was achieved. The chart illustrates the

expected trade-off: as the threshold increases, precision improves while recall declines. In Figure B6, the selected cutoff provides a more balanced classification rule for the pruned tree and supports a more reliable identification of hit songs than a purely default threshold choice.

After pruning and threshold tuning, the model was evaluated on both the training and testing sets to verify whether it could maintain similar performance outside the sample used for estimation. Looking across the main classification metrics, the gap between train and test results remains relatively small, which suggests that the simpler tree generalises more consistently than the initial unconstrained version.

At optimal cutoff (0.51):

	Train Set	Test Set	Difference (Test - Train)
ROC AUC	75.80%	70.05%	-5.75%
Accuracy	64.01%	60.79%	-3.22%
Precision	39.38%	36.40%	-2.98%
Recall	80.19%	75.05%	-5.14%
F1-Score	52.82%	49.03%	-3.80%

Figure B7: Train vs Test Performance of the Pruned Classification Tree

The results reported in Figure B7 support this interpretation. ROC AUC decreases from 75.80% in the training set to 70.05% in the test set, while recall moves from 80.19% to 75.05% and F1-score from 52.82% to 49.03%. These differences are limited enough to indicate that severe overfitting is no longer a major concern. Even so, the model's overall predictive quality remains only moderate, especially because precision stays relatively low, so the pruned Classification Tree appears more stable than the original version but still not strong enough to be considered the best final model.

To complement the metric comparison, the final pruned tree was also examined through its confusion matrix and classification report. This provides a more detailed view of how well the model distinguishes hit songs from non-hit songs and highlights the types of classification errors it makes most often.

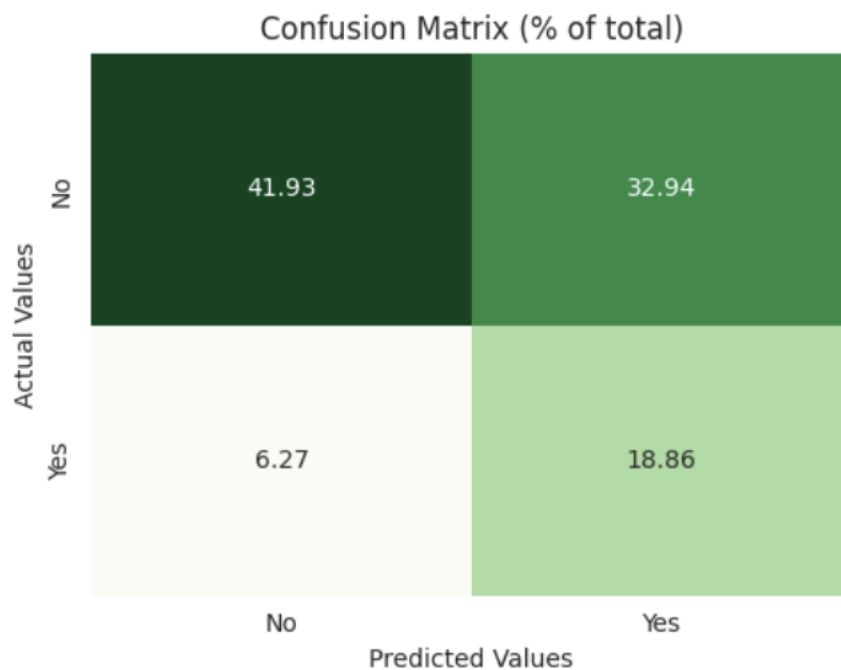


Figure B8: Confusion Matrix and Classification Report for the Pruned Classification Tree

A closer look at Figure B8 shows that the model correctly classifies 41.93% of all observations as non-hits and 18.86% as hits, while misclassifying 32.94% of observations as false positives and 6.27% as false negatives. This pattern is consistent with the model's objective of capturing hit songs, as reflected in the relatively high recall of 0.75 for the hit class. At the same time, the precision of 0.36 for hits remains low, meaning that many songs predicted as hits are actually non-hits. Overall, Figure B8 confirms that the pruned tree is better at identifying hits than avoiding false alarms, which limits its usefulness as a strong final classifier.

B2.3 Classification Model – Option: 2 Random Forest

This section introduces the Random Forest model as the second classification approach tested in the analysis. Since the initial Classification Tree showed limited predictive performance and a tendency to overfit, Random Forest was explored as a more robust alternative that combines multiple trees in order to improve stability and generalisation.

To implement this second approach, a Random Forest classifier was trained and tuned using cross-validation. The model was configured to address class imbalance and to search for an appropriate combination of tree depth, number of features considered at each split, and minimum leaf size. The following outputs present the selected parameters and the resulting predictive performance.

```

--- RECALL COMPARATIVE REPORT (at default cutoff) ---
1. Train Set (In-Bag):      74.6%
2. OOB Set (Internal Val):  71.9%
3. CV Score (GridSearch):   71.4%

Overfitting Gap (Train - OOB): 2.7%
>> Model is consistent and robust.

```

Figure B9: Recall Comparison and Overfitting Check for the Random Forest Model

The results indicate that the Random Forest model achieved very similar recall across the training, out-of-bag, and cross-validation sets, with values of **74.6%**, **71.9%**, and **71.4%**, respectively. The difference between the training and out-of-bag recall is only **2.7%**, suggesting that the model generalises well and does not show strong signs of overfitting. Overall, **Figure B9** suggests that the Random Forest provides a more stable and robust classification approach than the single Classification Tree.

To further refine the Random Forest model, the number of trees was examined by tracking how the out-of-bag recall changed as the forest size increased. This makes it possible to identify the point at which adding more trees no longer produces meaningful gains in predictive performance.

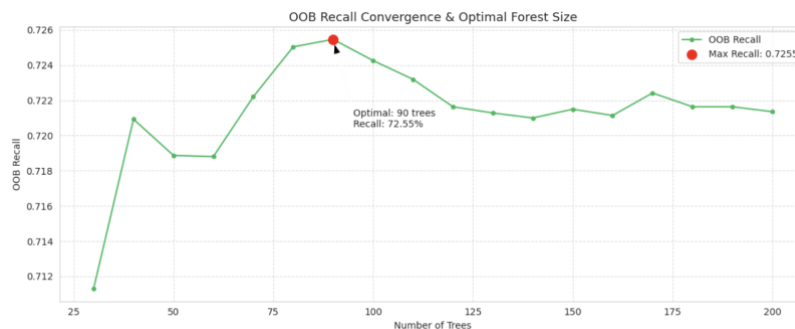


Figure B10: OOB Recall Convergence and Optimal Number of Trees for the Random Forest

The pattern in Figure B10 suggests that performance improves steadily at first and then stabilises as more trees are added. The highest out-of-bag recall was reached with 90 trees, where the model achieved a value of approximately 72.55%. Beyond this point, the gains become very small and the curve remains relatively flat, indicating that increasing the number of trees further would add complexity without delivering a meaningful improvement in recall.

After selecting the final Random Forest specification, the classification threshold was evaluated to determine whether the default cutoff should be adjusted. This was done by comparing precision and recall across different probability thresholds and selecting the value that maximized the F1-score.

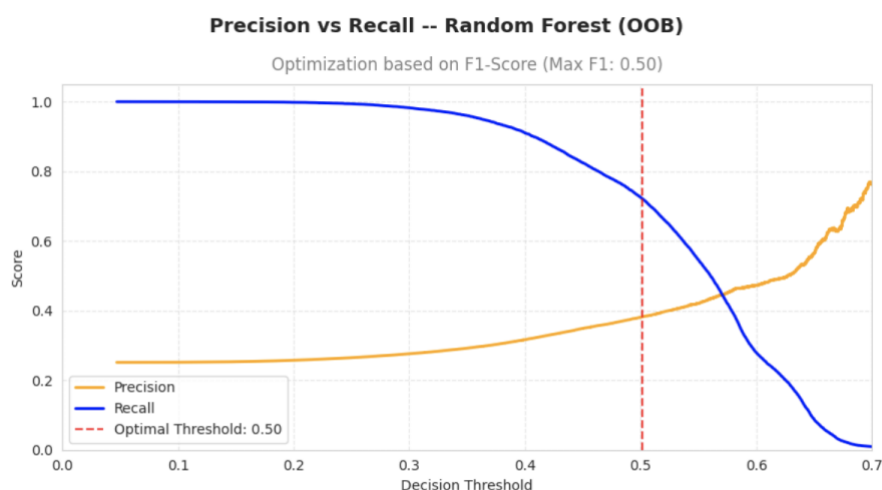


Figure B11: Precision-Recall Trade-off and Optimal Threshold for the Random Forest

In this case, the optimal threshold remained at 0.50, which means that the default classification cutoff already provided the best balance between precision and recall. The figure also illustrates the expected trade-off between the two metrics: higher thresholds increase precision but reduce recall. Overall, the Random Forest did not require an adjusted cutoff, since the standard threshold already offered the most suitable compromise under the F1-score criterion.

To evaluate the final Random Forest model, its performance was examined across the training and testing sets and then complemented with a confusion matrix, classification report, and ROC curve. Together, these outputs provide a fuller view of predictive quality, class-specific performance, and the model's ability to discriminate between hit and non-hit songs.

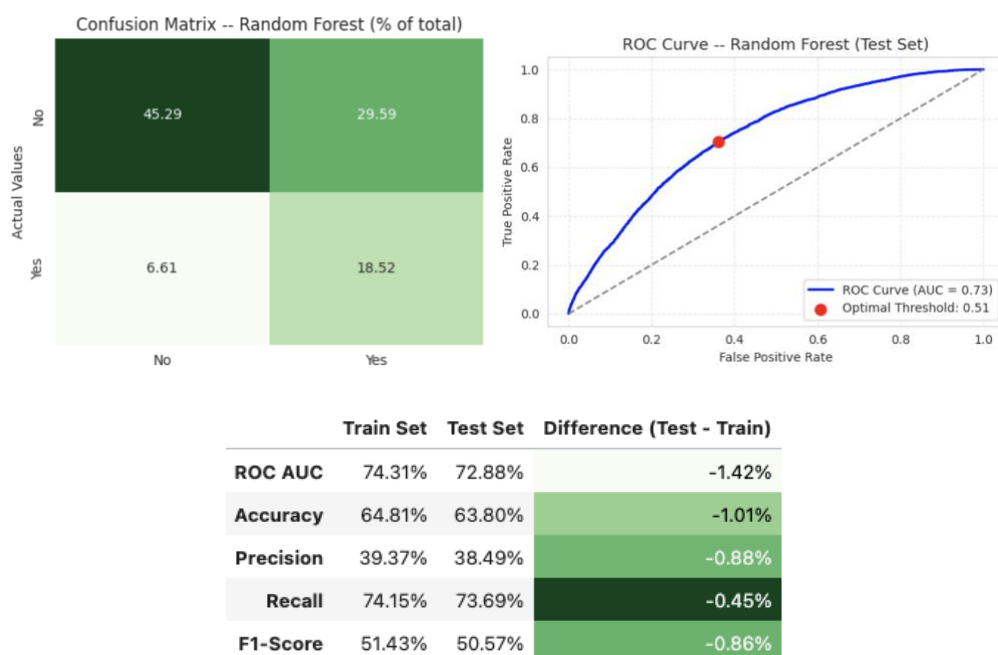


Figure B12: Final Performance Evaluation of the Random Forest Model

Overall, the results suggest that the Random Forest is stable and generalises well to unseen data. The gap between training and testing performance remains small across the main metrics, indicating no strong signs of overfitting. In addition, Figure B12 shows that the model achieves a test ROC AUC of 0.7288, with accuracy around 64%, recall of 0.74 for the hit class, and F1-score of 0.51. The confusion matrix also indicates that the model identifies a reasonable share of hit songs, although false positives remain present. Taken together, these results show that the Random Forest offers a more balanced and reliable performance than the previous Classification Tree.

B2.4 Classification Model - Option 3: Boosting

This section presents the third classification approach, based on a Boosting model. Unlike the previous methods, Boosting builds trees sequentially, with each new tree focusing on the errors made by the earlier ones. This approach was tested to determine whether a more adaptive ensemble method could deliver stronger predictive performance than both the single Classification Tree and the Random Forest model.

To implement the Boosting approach, an XGBoost classifier was trained and tuned using cross-validation. The model was configured to account for class imbalance through the `scale_pos_weight` parameter, while a grid search was used to identify the most suitable combination of learning rate, tree depth, sampling ratios, and minimum child weight.

```
--- RECALL COMPARATIVE REPORT (XGBoost) ---  
1. Train Set:  76.6%  
2. CV Score:   72.1%  
3. Test Set:   74.2%  
  
Overfitting Gap (Train - Test): 2.4%
```

Figure B13: Parameter Tuning and Recall Comparison for the XGBoost Model

The results suggest that the Boosting model achieved strong and consistent recall performance across the different evaluation stages. The best parameter combination included `max_depth = 5`, `learning_rate = 0.05`, `n_estimators = 200`, `subsample = 0.8`, `colsample_bytree = 0.8`, and `min_child_weight = 10`. In addition, Figure B13 shows that recall remained relatively stable across the training set (76.6%), cross-validation (72.1%), and test set (74.2%), with an overfitting gap of only 2.4%. Overall, this indicates that the XGBoost model is both robust and well-generalised.

To better understand how the Boosting model behaved as more trees were added, a learning curve was examined across different numbers of boosting rounds. This helps show whether performance improves steadily or begins to level off as the model becomes more complex.

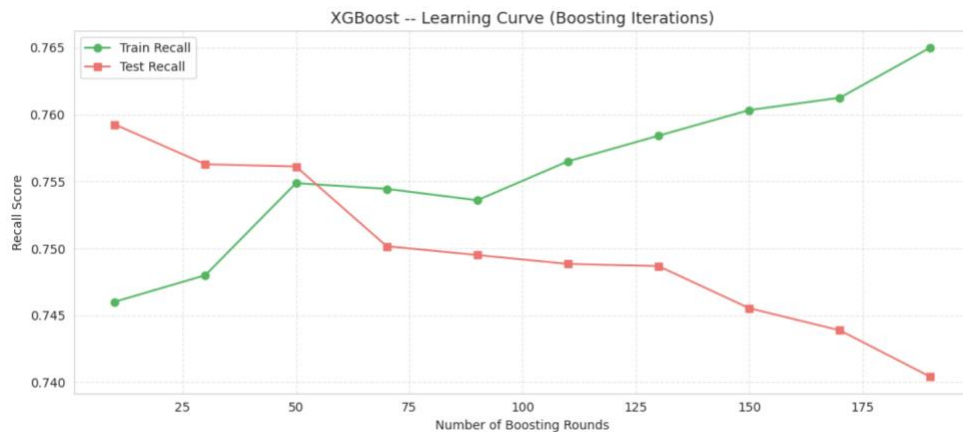


Figure B14: Learning Curve of the XGBoost Model Across Boosting Rounds

The pattern in Figure B14 shows that training recall gradually increases as the number of boosting rounds grows, while test recall remains relatively stable and then begins to decline slightly. This suggests that adding more boosting rounds improves the fit on the training data but does not necessarily translate into better out-of-sample performance. Overall, Figure B14 indicates that the selected number of rounds provides a reasonable balance, while also showing mild signs that excessive boosting could start to increase overfitting.

After fitting the final XGBoost model, the classification threshold was evaluated to determine whether the default cutoff should be adjusted. This was done by comparing precision and recall across different probability thresholds and identifying the value that maximized the F1-score.

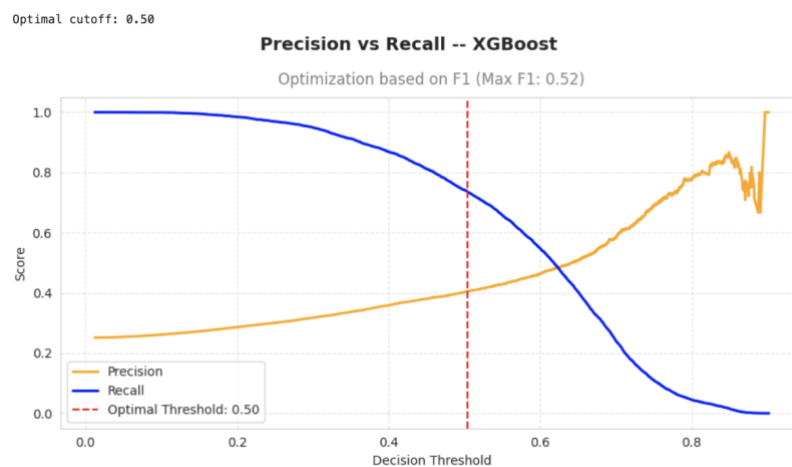


Figure B15: Precision-Recall Trade-off and Optimal Threshold for the XGBoost Model

Figure B15 indicates that the optimal threshold remained at 0.50, meaning that the default cutoff already provided the best balance between precision and recall. The curve also shows the expected trade-off between the two metrics, with precision increasing and recall decreasing as the threshold becomes more restrictive. Overall, the figure suggests that the final XGBoost model did not require a threshold adjustment, since the standard cutoff already delivered the strongest compromise under the F1-score criterion.

To assess the final XGBoost model, its performance was evaluated through a comparison of training and testing metrics, a confusion matrix with classification report, and the ROC curve. Together, these outputs provide a complete view of the model's generalisation ability, class-specific predictive performance, and overall discrimination capacity.

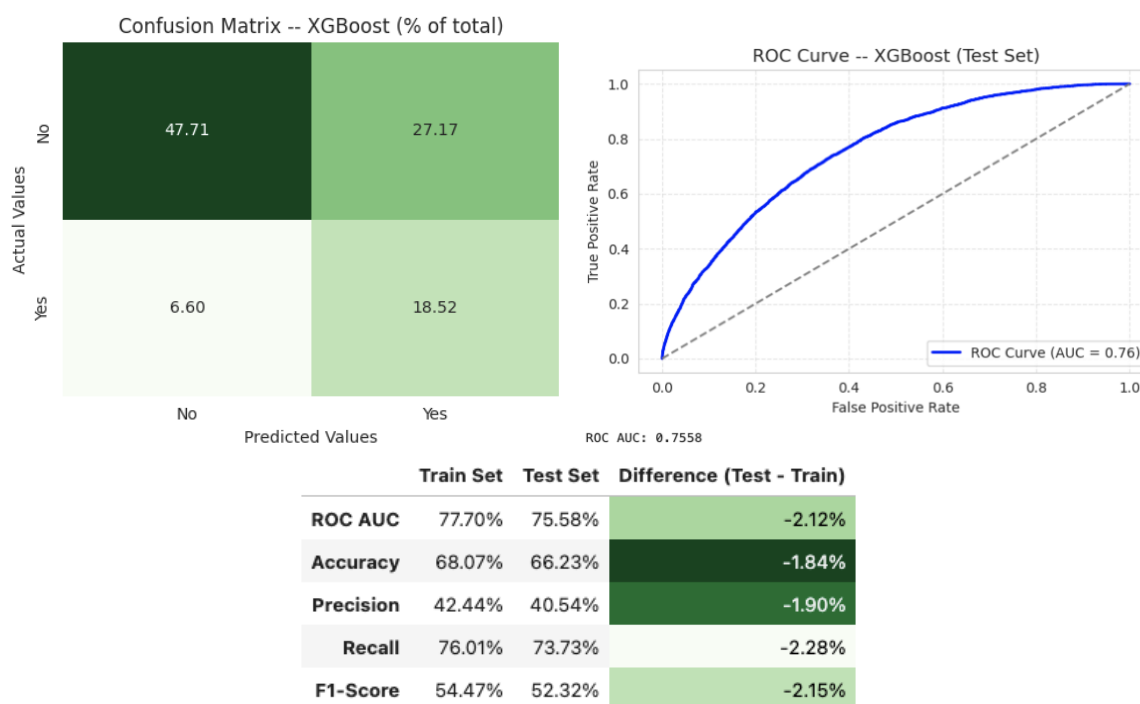


Figure B16: Final Performance Evaluation of the XGBoost Model

The results present in Figure B16, indicate that the XGBoost model is both stable and comparatively strong in predictive terms. The differences between training and testing metrics remain small, suggesting that overfitting is limited. In addition, the model achieves a test ROC AUC of 0.7558, accuracy of 66.23%, recall of 0.74 for the hit class, and an F1-score of 0.52. The confusion matrix also shows that the model correctly identifies a substantial share of both hit and non-hit songs, while keeping false negatives relatively low. Taken together, these results suggest that XGBoost provides the strongest overall balance between discrimination, stability, and classification performance among the models tested so far.

B2.5 Comparing the Classification Models and Final Selection

After evaluating each classification approach separately, the final step was to compare their results using a common set of performance criteria. Since the main business objective is to reduce the risk of missing truly successful songs, particular emphasis was placed on recall, while also considering model stability and overall predictive balance in the final selection.

To support the final selection, the three classification models were compared using the same evaluation metrics at their respective optimal thresholds. This allows their predictive performance to be assessed on a consistent basis and makes it easier to identify which model offers the strongest overall balance between recall, precision, and general classification quality.

FINAL MODEL COMPARISON (each model at optimal F1 threshold)
Cutoffs used – DT: 0.51 | RF: 0.50 | XGB: 0.50

Model	Cutoff	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	0.510	0.608	0.364	0.750	0.490	0.701
Random Forest	0.500	0.638	0.385	0.737	0.506	0.729
XGBoost	0.500	0.662	0.405	0.737	0.523	0.756

Figure B17: Final Comparison of the Classification Models at Their Optimal Thresholds

The comparison indicates that XGBoost provides the strongest overall performance among the three alternatives. While Decision Tree achieves the highest recall (0.750), it performs

worse on accuracy, precision, F1-score, and ROC-AUC, confirming its more limited predictive balance. Both ensemble methods outperform the single tree, but XGBoost stands out with the highest accuracy (0.662), precision (0.405), F1-score (0.523), and ROC-AUC (0.756), while maintaining the same recall (0.737) as the Random Forest. Overall, Figure B17 supports the selection of XGBoost as the final model, since it offers the best combination of stability, discrimination ability, and balanced classification performance.

To complement the numerical comparison, the final model selection was also visualised across the main evaluation metrics. This makes it easier to see where each model performs best and to identify which option offers the strongest overall balance for the business objective.

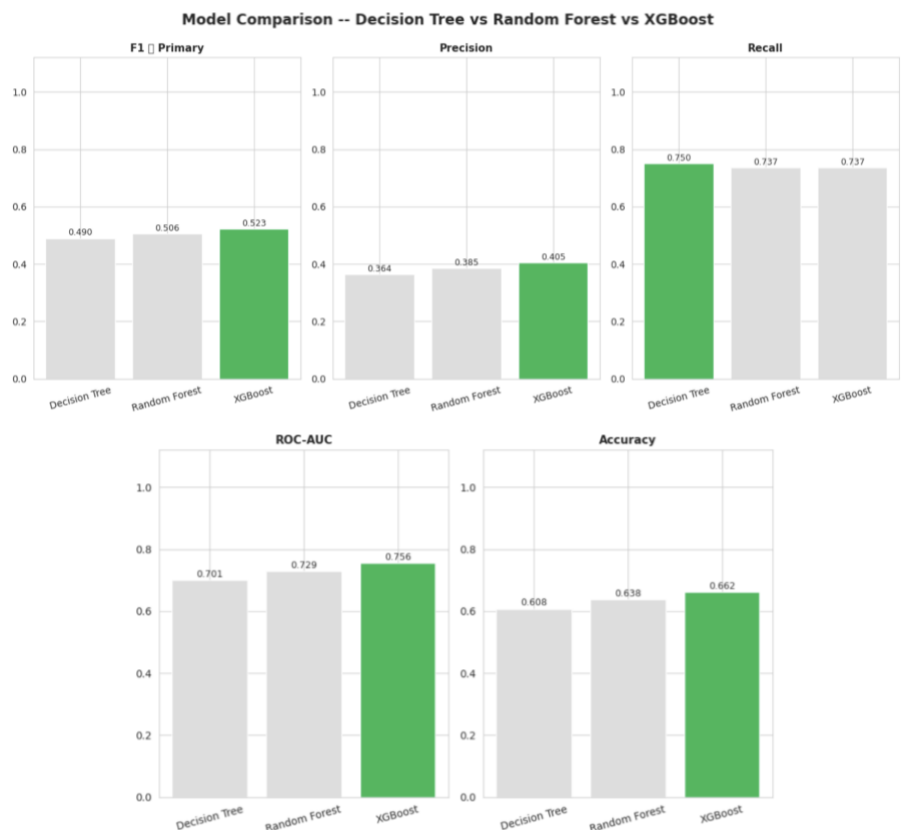


Figure B18: Visual Comparison of the Classification Models Across Key Metrics

The visual comparison reinforces the same conclusion as the summary table. XGBoost leads on F1-score, precision, ROC-AUC, and accuracy, while the Decision Tree performs best only in recall. Although recall was treated as the primary metric, the gap is small relative to the gains that XGBoost achieves on the other measures. As a result, Figure B18 supports the selection of XGBoost as the final model, since it offers the best overall trade-off between identifying hit songs and avoiding excessive false positives.

To further support the final model choice, the ROC curves of the three classifiers were plotted on the same graph. This allows their discrimination ability to be compared directly by showing how well each model separates hit songs from non-hit songs across all possible classification thresholds.

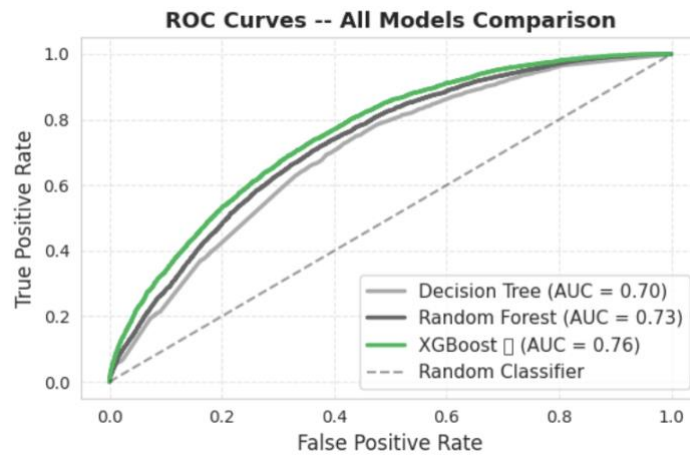


Figure B19: ROC Curve Comparison of the Classification Models

The combined ROC curves confirm that XGBoost delivers the strongest discrimination performance among the three models. Its curve lies consistently above those of the Decision Tree and Random Forest, and it also achieves the highest AUC, at approximately 0.76. The Random Forest follows with an AUC of 0.73, while the Decision Tree shows the weakest performance with an AUC of 0.70. Together, these results provide additional evidence that XGBoost is the most effective final model for the classification task.

TECHNICAL ANNEX C – BUSINESS INSIGHTS

This annex supports Section 3 of the main report and aligns with the technical Python notebook. It expands on the business insights from the final selected model.

Building on the final model selection, this annex focuses on translating predictive results into managerial insights. In particular, it examines which features contribute most to hit-song prediction, how these drivers can be interpreted from a business perspective, and how the model can support more profitable decision-making for record labels.

To better understand what drives the model's predictions, permutation importance was used to measure how much recall decreases when each feature is randomly disturbed. This provides a practical indication of which variables matter most for identifying potential hit songs.

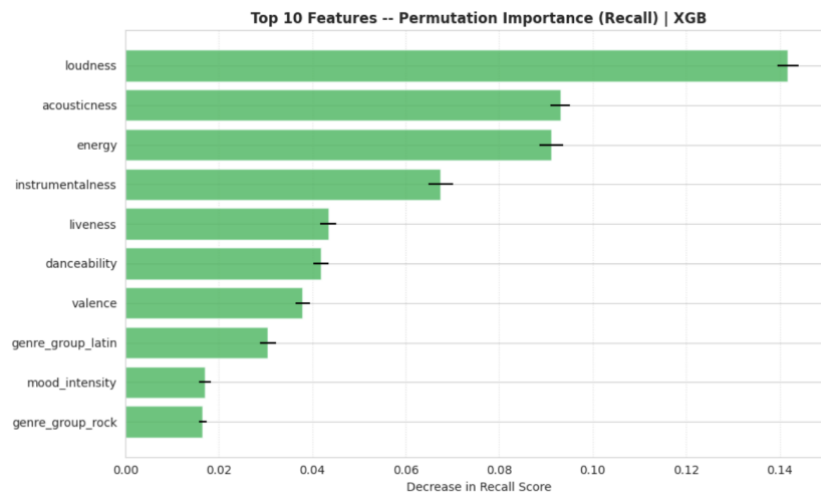


Figure C1: Top 10 Features by Permutation Importance for the XGBoost Model

Figure C1 presents the ranking of importance, it shows that loudness is the most influential predictor, followed by acousticness and energy. Other relevant variables include instrumentalness, liveness, and danceability, while genre-related variables appear lower in the ranking. Overall, this suggests that audio characteristics play a more important role than genre in predicting whether a song will become a hit, which is consistent with the main findings of the report.

To make the feature-importance results more concrete, the analysis also identified songs whose audio profiles most closely match the combination of characteristics associated with high hit potential. This helps translate the model's feature rankings into real song examples that reflect the patterns found in the data.

Top 5 Songs — audio-feature rank score (lower combined score = more hit-like)

	track_name	artists
#1	Naach Meri Rani	Guru Randhawa;Nikhita Gandhi;Tanishk Bagchi;Nora Fatehi
#2	Moving On	Lil Peep
#3	Thank You	JUNNY
#4	Turn Down for What (feat. Juicy J, 2 Chainz & French Montana) - Official Remix	DJ Snake;Lil Jon;Juicy J;2 Chainz;French Montana
#5	worthless	eli.

loudness	danceability	energy	acousticness	instrumentalness	liveness	valence	combined_rank_score
-5.1	0.676	0.488	0.061	0.0000	0.071	0.151	28232
-5.6	0.821	0.410	0.017	0.0000	0.113	0.197	28588
-8.4	0.876	0.473	0.012	0.0000	0.081	0.189	28821
-4.3	0.822	0.760	0.007	0.0000	0.040	0.362	29245
-4.2	0.675	0.530	0.016	0.0000	0.111	0.083	29509

Figure C2: Top 5 Songs with the Most Hit-Like Audio Feature Profiles

Figure C2 illustrates how the most hit-like songs tend to combine relatively high danceability, moderate-to-high energy, low instrumentalness, and low acousticness, while also showing stronger loudness levels. These patterns are consistent with the earlier importance analysis, reinforcing the idea that audio characteristics, rather than genre alone, are the main drivers of predicted hit potential.

To complement the permutation-based ranking, a SHAP-aligned composite score was used to identify songs whose overall audio profile is most strongly associated with hit potential. This provides a more interpretable view of how the key features combine in practice to produce songs that the model would consider especially promising.

Top 5 Songs — SHAP-aligned composite score (higher = better)

	track_name	loudness	danceability	valence	energy
#1	Ahora Te Puedes Marchar	-3.0	0.791	0.961	0.722
#2	What Makes You Beautiful	-2.5	0.726	0.888	0.787
#3	Beatbox	-4.2	0.857	0.773	0.700
#4	Gentleman	-3.3	0.878	0.619	0.723
#5	Strawberry Sunscreen	-3.9	0.688	0.909	0.678

acousticness	instrumentalness	liveness	mood_intensity	hit_score
0.068	0.0000	0.052	0.694	90.7
0.009	0.0000	0.060	0.699	90.3
0.041	0.0000	0.033	0.541	89.7
0.002	0.0000	0.052	0.448	89.6
0.036	0.0000	0.033	0.616	89.5

Figure C3: Top 5 Songs with the Highest SHAP-Aligned Hit Scores

Figure C3 shows examples that further reinforce the importance of audio characteristics in predicting success. The highest-scoring songs tend to show a combination of stronger loudness, high danceability, solid energy, very low instrumentalness, and low acousticness, alongside favourable values for valence and mood intensity. Overall, these examples support the view that commercially promising songs are defined more by their audio profile than by genre alone.

To translate the model into business terms, two profit-based decision scenarios were compared using the predicted probabilities from the final XGBoost model. This allows the classification threshold to be chosen not only on statistical grounds, but also according to the financial consequences of promoting non-hit songs versus missing true hits.

Scenario		Cost of Promoting Non-Hit (FP) \		
0	Opportunity-Focused	1000		
1	Budget-Control	4000		
		Cost of Missing Hit (FN)	Gain from Identifying Hit (TP) \	
0		5000	10000	
1		5000	10000	
		Optimal Threshold	Train Profit per Case	Test Profit per Case
0		0.279	1869.440	1823.650
1		0.447	601.890	476.590

Figure C4: Profit-Based Threshold Optimisation Under Alternative Business Scenarios

The comparison in Figure C4 shows that the optimal classification threshold depends strongly on the economic objective. Under the Opportunity-Focused scenario, where missing a true hit is much more costly than promoting a non-hit, the best threshold is lower (0.279), producing the highest expected profit per case on both the training and test sets (1869.44 and 1823.65, respectively). By contrast, the more conservative Budget-Control scenario leads to a higher threshold (0.447) and substantially lower expected profit. Overall, Figure C4 supports the idea that a more aggressive, high-recall strategy is economically preferable when the cost of missing a hit outweighs the cost of promoting a non-hit.

To further illustrate the economic implications of threshold choice, expected profit was plotted across the full range of decision thresholds for the two business scenarios. This makes it possible to see not only the optimal cutoff in each case, but also how sensitive financial outcomes are to more conservative or more aggressive promotion strategies.

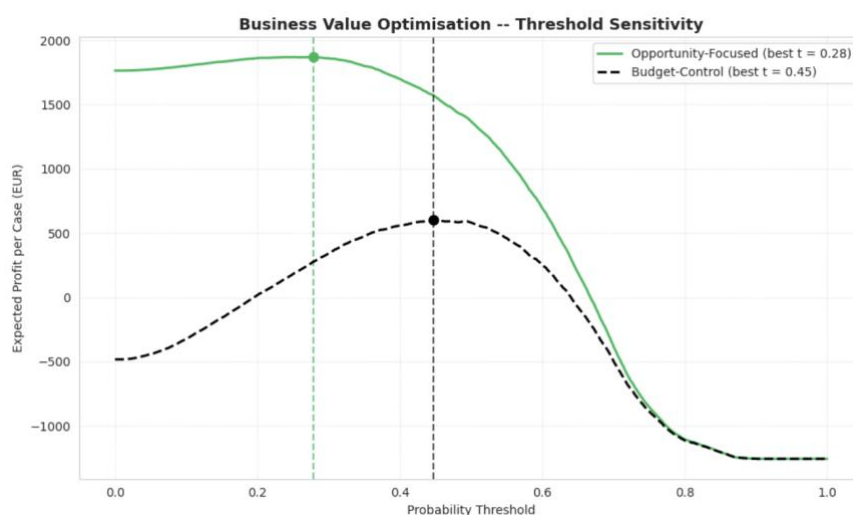


Figure C5: Threshold Sensitivity of Expected Profit Under Alternative Business Scenarios

The pattern in Figure C5 shows that the Opportunity-Focused scenario consistently generates higher expected profit and reaches its maximum at a lower threshold, around 0.28. In contrast, the Budget-Control scenario peaks at a higher threshold, around 0.45, but delivers a much lower maximum return. The figure also shows that profits fall sharply when the threshold becomes too high, since more true hits are missed. Overall, this visual evidence reinforces the conclusion that a lower, opportunity-oriented threshold is more profitable when missing a hit carries the greatest business cost.

To compare the practical consequences of the two business strategies, their confusion matrices were examined at the respective optimal thresholds. This makes it possible to see how each scenario balances the trade-off between capturing true hits and avoiding the promotion of non-hit songs.

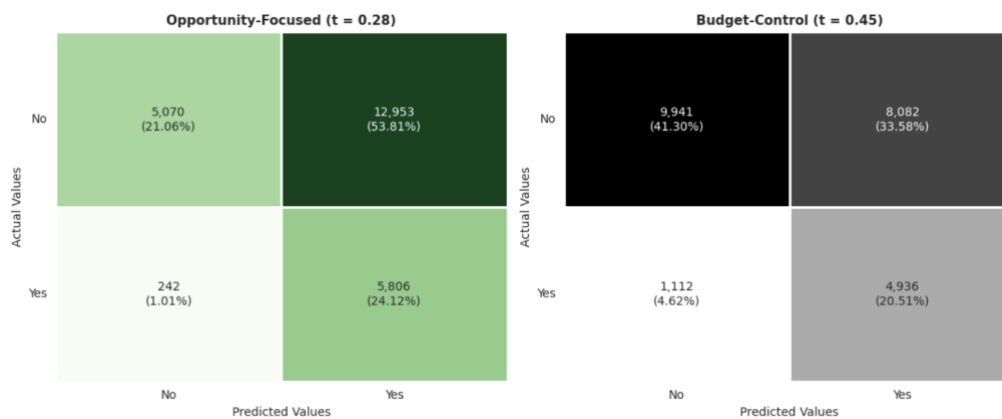


Figure C6: Confusion Matrices for the Opportunity-Focused and Budget-Control Scenarios

The contrast in Figure C6 makes the strategic trade-off very clear. Under the Opportunity-Focused scenario, the model captures almost all actual hits, with only 1.01% of observations classified as missed hits, but this comes with a much larger share of false positives (53.81%), meaning more non-hit songs would be promoted. In the Budget-Control scenario, false positives fall to 33.58%, but missed hits increase to 4.62%. Overall, Figure C6 shows that the opportunity-focused strategy is more aggressive and less efficient in the short run, but better aligned with the objective of avoiding costly missed hit songs.

